Table 1 - Video Format Timings — Detailed Timing Information

Field Rate[d]	VIC	Figure	Hactive	Vactive	I / P	Htotal	Hblank[d]	Hfront	Hsync	Hback	Hpol[v]	Vtotal	Vblank[d]	Vfront	Vsync	Vback	Vpol[v]	H Freq[d] (kHz)	V Freq (Hz)	Pixel Freq[d] (MHz)	Ln	Reference Standard	Notes
Low	60, 65	2	1280	720	Prog	3300	2020	1760	40	220	P	750	30	5	5	20	P	18.000	24.000[c]	59.400	1	SMPTE ST 296 [101]	d, f, x
	61, 66	2	1280	720	Prog	3960	2680	2420	40	220	P	750	30	5	5	20	P	18.750	25.000	74.250	1		d, f
	62, 67	2	1280	720	Prog	3300	2020	1760	40	220	P	750	30	5	5	20	P	22.500	30.000[c]	74.250	1		d, f
	108, 109	2	1280	720	Prog	2500	1220	960	40	220	P	750	30	5	5	20	P	36.000	48.000[c]	90.000	1		d, f, x
	79	2	1680	720	Prog	3300	1620	1360	40	220	P	750	30	5	5	20	P	18.000	24.000[c]	59.400	1		y
	80	2	1680	720	Prog	3168	1488	1228	40	220	P	750	30	5	5	20	P	18.750	25.000	59.400	1		y
	81	2	1680	720	Prog	2640	960	700	40	220	P	750	30	5	5	20	P	22.500	30.000[c]	59.400	1		y
	110	2	1680	720	Prog	2750	1070	810	40	220	P	750	30	5	5	20	P	36.000	48.000[c]	99.000	1		y
	32, 72	2	1920	1080	Prog	2750	830	638	44	148	P	1125	45	4	5	36	P	27.000	24.000[c]	74.250	1	SMPTE ST 274 [2]	r
	33, 73	2	1920	1080	Prog	2640	720	528	44	148	P	1125	45	4	5	36	P	28.125	25.000	74.250	1		r
	34, 74	2	1920	1080	Prog	2200	280	88	44	148	P	1125	45	4	5	36	P	33.750	30.000[c]	74.250	1		r
	111, 112	2	1920	1080	Prog	2750	830	638	44	148	P	1125	45	4	5	36	P	54.000	48.000[c]	148.500	1		r
	86	2	2560	1080	Prog	3750	1190	998	44	148	P	1100	20	4	5	11	P	26.400	24.000[c]	99.000	1		z
	87	2	2560	1080	Prog	3200	640	448	44	148	P	1125	45	4	5	36	P	28.125	25.000	90.000	1		z
	88	2	2560	1080	Prog	3520	960	768	44	148	P	1125	45	4	5	36	P	33.750	30.000[c]	118.800	1		z
	113	2	2560	1080	Prog	3750	1190	998	44	148	P	1100	20	4	5	11	P	52.800	48.000[c]	198.000	1		z
	93, 103	2	3840	2160	Prog	5500	1660	1276	88	296	P	2250	90	8	10	72	P	54.000	24.000[c]	297.000	1		d, f, aa
	94, 104	2	3840	2160	Prog	5280	1440	1056	88	296	P	2250	90	8	10	72	P	56.250	25.000	297.000	1		d, f, aa
	95, 105	2	3840	2160	Prog	4400	560	176	88	296	P	2250	90	8	10	72	P	67.500	30.000[c]	297.000	1		d, f, aa
	114, 116	2	3840	2160	Prog	5500	1660	1276	88	296	P	2250	90	8	10	72	P	108.000	48.000[c]	594.000	1		d, f, aa
	98	2	4096	2160	Prog	5500	1404	1020	88	296	P	2250	90	8	10	72	P	54.000	24.000[c]	297.000	1		d, f, bb
	99	2	4096	2160	Prog	5280	1184	968	88	128	P	2250	90	8	10	72	P	56.250	25.000	297.000	1		d, f, bb
	100	2	4096	2160	Prog	4400	304	88	88	128	P	2250	90	8	10	72	P	67.500	30.000[c]	297.000	1		d, f, bb
	115	2	4096	2160	Prog	5500	1404	1020	88	296	P	2250	90	8	10	72	P	108.000	48.000[c]	594.000	1		d, f, bb
	121	2	5120	2160	Prog	7500	2380	1996	88	296	P	2200	40	8	10	22	P	52.800	24.000[c]	396.000	1		cc
	122	2	5120	2160	Prog	7200	2080	1696	88	296	P	2200	40	8	10	22	P	55.000	25.000	396.000	1		cc
	123	2	5120	2160	Prog	6000	880	664	88	128	P	2200	40	8	10	22	P	66.000	30.000[c]	396.000	1		cc
	124	2	5120	2160	Prog	6250	1130	746	88	296	P	2475	315	8	10	297	P	118.800	48.000[c]	742.500	1		cc

Table 1 - Video Format Timings — Detailed Timing Information (continued)

Field Rate[d]	VIC	Figure	Hactive	Vactive	I / P	Htotal	Hblank[d]	Hfront	Hsync	Hback	Hpol[v]	Vtotal	Vblank[d]	Vfront	Vsync	Vback	Vpol[v]	H Freq[d] (kHz)	V Freq (Hz)	Pixel Freq[d] (MHz)	Ln	Reference Standard	Notes
Low (cont.)	194, 202	2	7680	4320	Prog	11000	3320	2552	176	592	P	4500	180	16	20	144	P	108.000	24.000[c]	1188.000	1	SMPTE ST 274 [2]	
	195, 203	2	7680	4320	Prog	10800	3120	2352	176	592	P	4400	80	16	20	44	P	110.000	25.000	1188.000	1		
	196, 204	2	7680	4320	Prog	9000	1320	552	176	592	P	4400	80	16	20	44	P	132.000	30.000[c]	1188.000	1		
	197, 205	2	7680	4320	Prog	11000	3320	2552	176	592	P	4500	180	16	20	144	P	216.000	48.000[c]	2376.000	1		
	210	2	10240	4320	Prog	12500	2260	1492	176	592	P	4950	630	16	20	594	P	118.800	24.000[c]	1485.000	1		dd
	211	2	10240	4320	Prog	13500	3260	2492	176	592	P	4400	80	16	20	44	P	110.000	25.000	1485.000	1		dd
	212	2	10240	4320	Prog	11000	760	288	176	296	P	4500	180	16	20	144	P	135.000	30.000[c]	1485.000	1		dd
	213	2	10240	4320	Prog	12500	2260	1492	176	592	P	4950	630	16	20	594	P	237.600	48.000[c]	2970.000	1		dd
50 Hz	17, 18	1	720	576	Prog	864	144	12	64	68	N	625	49	5	5	39	N	31.250	50.000	27.000	1	Rec. ITU-R BT.1358 [116]	
	21, 22	3	1440[b]	576	Int	1728[b]	288	24	126	138	N	625	24.5[a]	2	3	19	N	15.625	50.000	27.000	1	Rec. ITU-R BT.656 [114]	j, s
	23, 24	1	1440[b]	288	Prog	1728[b]	288	24	126	138	N	312	24	2	3	19	N	15.625	50.080	27.000	1	Rec. ITU-R BT.1358 [116]	k, r, s, w
												313	25	3					49.920				
												314	26	4					49.761				
	29, 30	1	1440[b]	576	Prog	1728[b]	288	24	128	136	N	625	49	5	5	39	N	31.250	50.000	54.000	1		m, n, r
	25, 26	3	2880[b]	576	Int	3456[b]	576	48	252	276	N	625	24.5[a]	2	3	19	N	15.625	50.000	54.000	1	Rec. ITU-R BT.656 [114]	l, q, r, u
	27, 28	1	2880[b]	288	Prog	3456[b]	576	48	252	276	N	312	24	2	3	19	N	15.625	50.080	54.000	1		k, l, p, q, u, w
												313	25	3					49.920				
												314	26	4					49.761				
	37, 38	1	2880[b]	576	Prog	3456[b]	576	48	256	272	N	625	49	5	5	39	N	31.250	50.000	108.000	1	Rec. ITU-R BT.1358 [116]	m, o
	19, 68	2	1280	720	Prog	1980	700	440	40	220	P	750	30	5	5	20	P	37.500	50.000	74.250	1	SMPTE ST 296 [101]	d, f
	82	2	1680	720	Prog	2200	520	260	40	220	P	750	30	5	5	20	P	37.500	50.000	82.500	1		d, f, y
	39	5	1920	1080	Int	2304	384	32	168	184	P	1250	85	23	5	57	N	31.250	50.000	72.000	1	AS 4933.1-2005 [126]	i
	20	4	1920	1080	Int	2640	720	528	44	148	P	1125	22.5[a]	2	5	15	P	28.125	50.000	74.250	1	SMPTE ST 274 [2]	d, f
	31, 75	2	1920	1080	Prog	2640	720	528	44	148	P	1125	45	4	5	36	P	56.250	50.000	148.500	1		r
	89	2	2560	1080	Prog	3300	740	548	44	148	P	1125	45	4	5	36	P	56.250	50.000	185.625	1		r, z
	96, 106	2	3840	2160	Prog	5280	1440	1056	88	296	P	2250	90	8	10	72	P	112.500	50.000	594.000	1		d, f, aa
	101	2	4096	2160	Prog	5280	1184	968	88	128	P	2250	90	8	10	72	P	112.500	50.000	594.000	1		d, f, bb
	125	2	5120	2160	Prog	6600	1480	1096	88	296	P	2250	90	8	10	72	P	112.500	50.000	742.500	1		cc
	198, 206	2	7680	4320	Prog	10800	3120	2352	176	592	P	4400	80	16	20	44	P	220.000	50.000	2376.000	1		
	214	2	10240	4320	Prog	13500	3260	2492	176	592	P	4400	80	16	20	44	P	220.000	50.000	2970.000	1		dd

Table 1 - Video Format Timings — Detailed Timing Information (continued)

Field Rate[d]	VIC	Figure	Hactive	Vactive	I / P	Htotal	Hblank[d]	Hfront	Hsync	Hback	Hpol[v]	Vtotal	Vblank[d]	Vfront	Vsync	Vback	Vpol[v]	H Freq[d] (kHz)	V Freq (Hz)	Pixel Freq[d] (MHz)	Ln	Reference Standard	Notes
60 Hz[c]	1	1	640	480	Prog	800	160	16	96	48	N	525	45	10	2	33	N	31.469	59.940[c]	25.175	1	VESA DMT [124]	d, h
	2, 3	1	720	480	Prog	858	138	16	62	60	N	525	45	9	6	30	N	31.469	59.940[c]	27.000	7	CTA-770.2 [30]	f
	6, 7	3	1440[b]	480	Int	1716[b]	276	38	124	114	N	525	22.5[a]	4	3	15	N	15.734	59.940[c]	27.000	4		f, s
	8, 9	1	1440[b]	240	Prog	1716[b]	276	38	124	114	N	262	22	4	3	15	N	15.734	60.054[c]	27.000	4		k, r, s, u, w
												263	23	5					59.826[c]				
	14, 15	1	1440[b]	480	Prog	1716[b]	276	32	124	120	N	525	45	9	6	30	N	31.469	59.940[c]	54.000	7		m, n, q, r
	10, 11	3	2880[b]	480	Int	3432[b]	552	76	248	228	N	525	22.5[a]	4	3	15	N	15.734	59.940[c]	54.000	4		l, q, u
	12, 13	1	2880[b]	240	Prog	3432[b]	552	76	248	228	N	262	22	4	3	15	N	15.734	60.054[c]	54.000	4		k, l, q, u, w
												263	23	5					59.826[c]				
	35, 36	1	2880[b]	480	Prog	3432[b]	552	64	248	240	N	525	45	9	6	30	N	31.469	59.940[c]	108.000	7		m, o
	4, 69	2	1280	720	Prog	1650	370	110	40	220	P	750	30	5	5	20	P	45.000	60.000[c]	74.250	1	CTA-770.3 [31]	d, f
	83	2	1680	720	Prog	2200	520	260	40	220	P	750	30	5	5	20	P	45.000	60.000[c]	99.000	1		d, f, y
	5	4	1920	1080	Int	2200	280	88	44	148	P	1125	22.5[a]	2	5	15	P	33.750	60.000[c]	74.250	1		d, f
	16, 76	2	1920	1080	Prog	2200	280	88	44	148	P	1125	45	4	5	36	P	67.500	60.000[c]	148.500	1	SMPTE ST 274 [2]	r
	90	2	2560	1080	Prog	3000	440	248	44	148	P	1100	20	4	5	11	P	66.000	60.000[c]	198.000	1		r, z
	97, 107	2	3840	2160	Prog	4400	560	176	88	296	P	2250	90	8	10	72	P	135.000	60.000[c]	594.000	1		d, f, aa
	102	2	4096	2160	Prog	4400	304	88	88	128	P	2250	90	8	10	72	P	135.000	60.000[c]	594.000	1		d, f, bb
	126	2	5120	2160	Prog	5500	380	164	88	128	P	2250	90	8	10	72	P	135.000	60.000[c]	742.500	1		cc
	199, 207	2	7680	4320	Prog	9000	1320	552	176	592	P	4400	80	16	20	44	P	264.000	60.000[c]	2376.000	1		
	215	2	10240	4320	Prog	11000	760	288	176	296	P	4500	180	16	20	144	P	270.000	60.000[c]	2970.000	1		dd
100 Hz	42, 43	1	720	576	Prog	864	144	12	64	68	N	625	49	5	5	39	N	62.500	100.000	54.000	1	Rec. ITU-R BT.1358 [116]	
	44, 45	3	1440[b]	576	Int	1728[b]	288	24	126	138	N	625	24.5[a]	2	3	19	N	31.250	100.000	54.000	1	Rec. ITU-R BT.656 [114]	t
	41, 70	2	1280	720	Prog	1980	700	440	40	220	P	750	30	5	5	20	P	75.000	100.000	148.500	1	SMPTE ST 296 [101]	
	84	2	1680	720	Prog	2000	320	60	40	220	P	825	105	5	5	95	P	82.500	100.000	165.000	1		y
	40	4	1920	1080	Int	2640	720	528	44	148	P	1125	22.5[a]	2	5	15	P	56.250	100.000	148.500	1	SMPTE ST 274 [2]	
	64, 77	2	1920	1080	Prog	2640	720	528	44	148	P	1125	45	4	5	36	P	112.500	100.000	297.000	1		
	91	2	2560	1080	Prog	2970	410	218	44	148	P	1250	170	4	5	161	P	125.000	100.000	371.250	1		z
	117, 119	2	3840	2160	Prog	5280	1440	1056	88	296	P	2250	90	8	10	72	P	225.000	100.000	1188.000	1		
	218	2	4096	2160	Prog	5280	1184	800	88	296	P	2250	90	8	10	72	P	225.000	100.000	1188.000	1		
	127	2	5120	2160	Prog	6600	1480	1096	88	296	P	2250	90	8	10	72	P	225.000	100.000	1485.000	1		cc
	200, 208	2	7680	4320	Prog	10560	2880	2112	176	592	P	4500	180	16	20	144	P	450.000	100.000	4752.000	1		
	216	2	10240	4320	Prog	13200	2960	2192	176	592	P	4500	180	16	20	144	P	450.000	100.000	5940.000	1		dd

Table 1 - Video Format Timings — Detailed Timing Information (continued)

Field Rate[d]	VIC	Figure	Hactive	Vactive	I / P	Htotal	Hblank[d]	Hfront	Hsync	Hback	Hpol[v]	Vtotal	Vblank[d]	Vfront	Vsync	Vback	Vpol[v]	H Freq[d] (kHz)	V Freq (Hz)	Pixel Freq[d] (MHz)	Ln	Reference Standard	Notes
120 Hz[c]	48, 49	1	720	480	Prog	858	138	16	62	60	N	525	45	9	6	30	N	62.937	119.880[c]	54.000	7	CTA-770.2 [30]	
	50, 51	3	1440[b]	480	Int	1716[b]	276	38	124	114	N	525	22.5[a]	4	3	15	N	31.469	119.880[c]	54.000	4		t
	47, 71	2	1280	720	Prog	1650	370	110	40	220	P	750	30	5	5	20	P	90.000	120.000[c]	148.500	1	SMPTE ST 296 [101]	
	85	2	1680	720	Prog	2000	320	60	40	220	P	825	105	5	5	95	P	99.000	120.000[c]	198.000	1		y
	46	4	1920	1080	Int	2200	280	88	44	148	P	1125	22.5[a]	2	5	15	P	67.500	120.000[c]	148.500	1	SMPTE ST 274 [2]	
	63, 78	2	1920	1080	Prog	2200	280	88	44	148	P	1125	45	4	5	36	P	135.000	120.000[c]	297.000	1		
	92	2	2560	1080	Prog	3300	740	548	44	148	P	1250	170	4	5	161	P	150.000	120.000[c]	495.000	1		z
	118, 120	2	3840	2160	Prog	4400	560	176	88	296	P	2250	90	8	10	72	P	270.000	120.000[c]	1188.000	1		
	219	2	4096	2160	Prog	4400	304	88	88	128	P	2250	90	8	10	72	P	270.000	120.000[c]	1188.000	1		
	193	2	5120	2160	Prog	5500	380	164	88	128	P	2250	90	8	10	72	P	270.000	120.000[c]	1485.000	1		cc
	201, 209	2	7680	4320	Prog	8800	1120	352	176	592	P	4500	180	16	20	144	P	540.000	120.000[c]	4752.000	1		
	217	2	10240	4320	Prog	11000	760	288	176	296	P	4500	180	16	20	144	P	540.000	120.000[c]	5940.000	1		dd
200 Hz	52, 53	1	720	576	Prog	864	144	12	64	68	N	625	49	5	5	39	N	125.000	200.000	108.000	1	Rec. ITU-R BT.1358 [116]	
	54, 55	3	1440[b]	576	Int	1728[b]	288	24	126	138	N	625	24.5[a]	2	3	19	N	62.500	200.000	108.000	1	Rec. ITU-R BT.656 [114]	t
240 Hz[c]	56, 57	1	720	480	Prog	858	138	16	62	60	N	525	45	9	6	30	N	125.874	239.760[c]	108.000	7	CTA-770.2 [30]	
	58, 59	3	1440[b]	480	Int	1716[b]	276	38	124	114	N	525	22.5[a]	4	3	15	N	62.937	239.760[c]	108.000	4		t

Notes:

a. Fractional Vblank values indicate that the number of Blanking Lines varies (see timing diagram for more details).

b. The pixels for the 720(1440)x480i@59.94/60 Hz, 720(1440)x240p@59.94/60 Hz, 720(1440)x576i@50 Hz, and 720(1440)x288p@50 Hz Video Formats are double clocked to meet minimum speed requirements of the interface, thus H active is shown as 1440, instead of 720. At higher field rates, these formats continue to be double clocked — even though double clocking is unnecessary. Each pixel of the 1440xN 480p and 576p formats, as well as the 2880xN 480i, 240p, 480p, 576i, 288p, and 576p formats, is repeated a variable number of times. The repeat value is communicated using the AVI InfoFrames (see Section 6.4).

c. The vertical frequencies of the 240p, 480p, and 480i Video Formats are tuned by a factor of exactly 1000/1001 for NTSC video compatibility, while the 576p, 576i, and the HDTV Video Formats are typically not adjusted. The VESA DMT standard [124] specifies a ± 0.5% pixel clock frequency tolerance. Therefore, the nominally 25.175 MHz pixel clock frequency value given for Video Identification Code 1 may be adjusted to 25.2 MHz to obtain an exact 60 Hz vertical frequency.

d. Data in this column is provided for informational purposes only.